

PRIMARY ARRAY OPERATIONS

FRANCIS REUBEN R SECTION B RVUN25CSE408

PROGRAM 1: Input and Display Array Elements

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

#include <stdio.h>

```
int main() {  
    int n, i;  
    printf("Enter size: ");  
    scanf("%d", &n);  
    int a[n];  
    printf("Enter %d elements: ", n);  
    for(i = 0; i < n; i++) scanf("%d", &a[i]);  
    printf("Array: ");  
    for(i = 0; i < n; i++) printf("%d ", a[i]);  
    printf("\n");  
    return 0;  
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog1  
Enter size: 5  
Enter 5 elements: 1 2 3 -1 -5  
Array: 1 2 3 -1 -5  
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog1  
Enter size: 6  
Enter 6 elements: 12 -99 510 89 1200 -8889  
Array: 12 -99 510 89 1200 -8889  
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog1  
Enter size: 8  
Enter 8 elements: -1 15 -166 18 199 2000 -999 -15  
Array: -1 15 -166 18 199 2000 -999 -15  
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```

PROGRAM 2: Sum of Array Elements

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i, sum = 0;
```

```
    printf("Enter size: ");
```

```
    scanf("%d", &n);
```

```
    int a[n];
```

```
    printf("Enter %d elements: ", n);
```

```
    for(i = 0; i < n; i++) {
```

```
        scanf("%d", &a[i]);
```

```
        sum += a[i];}
```

```
    printf("Sum: %d\n", sum);
```

```
    printf("\n");
```

```
    return 0;
```

```
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog2
```

```
Enter size: 5
```

```
Enter 5 elements: -1 2 -12 -5 0
```

```
Sum: -16
```

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog2
```

```
Enter size: 8
```

```
Enter 8 elements: 12 100 56 -8 999 10 2 2
```

```
Sum: 1173
```

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog2
```

```
Enter size: 2
```

```
Enter 2 elements: 1 -1
```

```
Sum: 0
```

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```

PROGRAM 3: Average of Array Elements

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i, sum = 0;
```

```
    printf("Enter size: ");
```

```
    scanf("%d", &n);
```

```
    int a[n];
```

```
    printf("Enter %d elements: ", n);
```

```
    for(i = 0; i < n; i++) {
```

```
        scanf("%d", &a[i]);
```

```
        sum += a[i];
```

```
    }
```

```
    printf("Average: %.2f\n", (float)sum/n);
```

```
    printf("\n");
```

```
    return 0;
```

```
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog3
```

```
Enter size: 5
```

```
Enter 5 elements: 12 -12 99 10 20
```

```
Average: 25.80
```

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog3
```

```
Enter size: 4
```

```
Enter 4 elements: 1 2 3 4
```

```
Average: 2.50
```

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog3
```

```
Enter size: 6
```

```
Enter 6 elements: -99 -99 -99 1 1 1
```

```
Average: -49.00
```

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ █
```

PROGRAM 4: Largest Element in an Array

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i, max;
```

```
    printf("Enter size: ");
```

```
    scanf("%d", &n);
```

```
    int a[n];
```

```
    printf("Enter %d elements: ", n);
```

```
    scanf("%d", &a[0]);
```

```
    max = a[0];
```

```
    for(i = 1; i < n; i++) {
```

```
        scanf("%d", &a[i]);
```

```
        if(a[i] > max) max = a[i];
```

```
    }
```

```
    printf("Largest: %d\n", max);
```

```
    printf("\n");
```

```
    return 0;
```

```
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog4
```

```
Enter size:
```

```
5
```

```
Enter 5 elements:  1 2 3 -99 99
```

```
Largest: 99
```

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog4
```

```
Enter size: 3
```

```
Enter 3 elements:  1 2 3
```

```
Largest: 3
```

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog4
```

```
Enter size: 6
```

```
Enter 6 elements: -99 50 66 90 122 -999
```

```
Largest: 122
```

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```

PROGRAM 5: Smallest Element in an Array

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i, min;
```

```
    printf("Enter size: ");
```

```
    scanf("%d", &n);
```

```
    int a[n];
```

```
    printf("Enter %d elements: ", n);
```

```
    scanf("%d", &a[0]);
```

```
    min = a[0];
```

```
    for(i = 1; i < n; i++) {
```

```
        scanf("%d", &a[i]);
```

```
        if(a[i] < min) min = a[i];
```

```
    }
```

```
    printf("Smallest: %d\n", min);
```

```
    printf("\n");
```

```
    return 0;
```

```
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog5
```

```
Enter size: 5
```

```
Enter 5 elements:  1 2 3 90 99
```

```
Smallest: 1
```

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog5
```

```
Enter size: 3
```

```
Enter 3 elements: 100 -99 918
```

```
Smallest: -99
```

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog5
```

```
Enter size: 6
```

```
Enter 6 elements: 100 2030 -300 -979 747 600
```

```
Smallest: -979
```

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```

PROGRAM 6: Count Positive, Negative, and Zero Elements

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i, pos = 0, neg = 0, zero = 0;
```

```
    printf("Enter size: ");
```

```
    scanf("%d", &n);
```

```
    int a[n];
```

```
    printf("Enter %d elements: ", n);
```

```
    for(i = 0; i < n; i++) {
```

```
        scanf("%d", &a[i]);
```

```
        if(a[i] > 0) pos++;
```

```
        else if(a[i] < 0) neg++;
```

```
        else zero++;
```

```
    }
```

```
    printf("Positive: %d, Negative: %d, Zero: %d\n", pos, neg, zero);
```

```
    return 0;
```

```
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog6
Enter size: 3
Enter 3 elements: 1 2 -099
Positive: 2, Negative: 1, Zero: 0
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog6
Enter size: 5
Enter 5 elements: 1 -2 -44 -99 100
Positive: 2, Negative: 3, Zero: 0
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog6
Enter size: 10
Enter 10 elements: 12 23 -24 0 0 15 18 99 100 -99
Positive: 6, Negative: 2, Zero: 2
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```

PROGRAM 7: Count Even and Odd Numbers

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i, even = 0, odd = 0;
```

```
    printf("Enter size: ");
```

```
    scanf("%d", &n);
```

```
    int a[n];
```

```
    printf("Enter %d elements: ", n);
```

```
    for(i = 0; i < n; i++) {
```

```
        scanf("%d", &a[i]);
```

```
        if(a[i] % 2 == 0) even++;
```

```
        else odd++;
```

```
    }
```

```
    printf("Even: %d, Odd: %d\n", even, odd);
```

```
    return 0;
```

```
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog7
Enter size: 5
Enter 5 elements: 1 2 3 -6 -7
Even: 2, Odd: 3
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog7
Enter size: 3
Enter 3 elements: -999 10 5
Even: 1, Odd: 2
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog7
Enter size: 6
Enter 6 elements: 1 2 3 15 6 18
Even: 3, Odd: 3
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```

PROGRAM 8: Search for a Given Element (Linear Search)

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {
    int n, i, key, found = 0;
    printf("Enter size: ");
    scanf("%d", &n);
    int a[n];
    printf("Enter %d elements: ", n);
    for(i = 0; i < n; i++) scanf("%d", &a[i]);
    printf("Enter element to search: ");
    scanf("%d", &key);
    for(i = 0; i < n; i++) {
        if(a[i] == key) {
            printf("Found at index %d\n", i);
            found = 1;
            break;
        }
    }
    if(!found) printf("Not found\n");
    return 0;
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog8
Enter size: 5
Enter 5 elements: 19 125 -99 100 12
Enter element to search: -99
Found at index 2
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog8
Enter size: 10
Enter 10 elements: 1 2 15 16 18 99 10 -99 -999
1
Enter element to search: 0
Not found
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog8
Enter size: 4
Enter 4 elements: 1 -88 102 -500
Enter element to search: 102
Found at index 2
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```


PROGRAM 9: Find Frequency of Each Element

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {
    int n, i, j, count;
    printf("Enter size: ");
    scanf("%d", &n);
    int a[n], visited[n];
    printf("Enter %d elements: ", n);
    for(i = 0; i < n; i++) {
        scanf("%d", &a[i]);
        visited[i] = 0;
    }
    printf("Frequency:\n");
    for(i = 0; i < n; i++) {
        if(visited[i]) continue;
        count = 1;
        for(j = i + 1; j < n; j++) {
            if(a[i] == a[j]) {
                count++;
                visited[j] = 1;
            }
        }
        printf("%d occurs %d time(s)\n", a[i], count);
    }
    return 0;
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog9
Enter size: 5
Enter 5 elements: 1 2 3 10 19
Frequency:
1 occurs 1 time(s)
2 occurs 1 time(s)
3 occurs 1 time(s)
10 occurs 1 time(s)
19 occurs 1 time(s)
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog9
Enter size: 10
Enter 10 elements: 1 1 25 60 12 12 -99 -91 10 0
Frequency:
1 occurs 2 time(s)
25 occurs 1 time(s)
60 occurs 1 time(s)
12 occurs 2 time(s)
-99 occurs 1 time(s)
-91 occurs 1 time(s)
10 occurs 1 time(s)
0 occurs 1 time(s)
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog9
Enter size: 5
Enter 5 elements: 10 10 10 0 10 0
Frequency:
10 occurs 4 time(s)
0 occurs 1 time(s)
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```

PROGRAM 10: Reverse the Array

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {
    int n, i, j, count;
    printf("Enter size: ");
    scanf("%d", &n);
    int a[n], visited[n];
    printf("Enter %d elements: ", n);
    for(i = 0; i < n; i++) {
        scanf("%d", &a[i]);
        visited[i] = 0;}
    printf("Frequency:\n");
    for(i = 0; i < n; i++) {
        if(visited[i]) continue;
        count = 1;
        for(j = i + 1; j < n; j++) {
            if(a[i] == a[j]) {
                count++;
                visited[j] = 1;}}
        printf("%d occurs %d time(s)\n", a[i], count);
    }
    return 0;
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog10
Enter size: 3
Enter 3 elements:  1 2 3
Reversed: 3 2 1
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog10
Enter size: 5
Enter 5 elements: 10 -99 0 99 -10
Reversed: -10 99 0 -99 10
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog10
Enter size: 10 50 50 10 50
Enter 10 elements: 10 50 50 10 50 10 50 50 10 50
Reversed: 10 50 10 50 50 10 50 10 50 50
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```

PROGRAM 11: Copy One Array to Another

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {
    int n, i;
    printf("Enter size: ");
    scanf("%d", &n);
    int a[n], b[n];
    printf("Enter %d elements: ", n);
    for(i = 0; i < n; i++) {
        scanf("%d", &a[i]);
        b[i] = a[i];
    }
    printf("Copied array: ");
    for(i = 0; i < n; i++) printf("%d ", b[i]);
    printf("\n");
    return 0;
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog11
Enter size: 5
Enter 5 elements: 1 2 3 10 60
Copied array: 1 2 3 10 60
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog11
Enter size: 6
Enter 6 elements: -99 10 15 60 12
1
Copied array: -99 10 15 60 12 1
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog11
Enter size: 8
Enter 8 elements: 10 40 50 -60 -70 -10 -99 19
Copied array: 10 40 50 -60 -70 -10 -99 19
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```

PROGRAM 12: Insert an Element at a Specific Position

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {
    int n, i, pos, val;
    printf("Enter size: ");
    scanf("%d", &n);
    int a[n+1];
    printf("Enter %d elements: ", n);
    for(i = 0; i < n; i++) scanf("%d", &a[i]);
    printf("Enter position and value: ");
    scanf("%d %d", &pos, &val);
    for(i = n; i > pos; i--) a[i] = a[i-1];
    a[pos] = val;
    printf("After insertion: ");
    for(i = 0; i <= n; i++) printf("%d ", a[i]);
    printf("\n");
    return 0;
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog12
Enter size: 5
Enter 5 elements: 10 50 69 10 67
Enter position and value: 2 20
After insertion: 10 50 20 69 10 67
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog12
Enter size: 6
Enter 6 elements: 1 2 60 -99 10 12
Enter position and value: 10 20
After insertion: 1 2 60 -99 10 12 4
Segmentation fault
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog12
Enter size: 10
Enter 10 elements: -99 10 394 5873 239 2192 58 939 -99 -175
Enter position and value: 5 -999999
After insertion: -99 10 394 5873 239 -999999 2192 58 939 -99 -175
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```

PROGRAM 13: Delete an Element from a Specific Position

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {  
    int n, i, pos;  
    printf("Enter size: ");  
    scanf("%d", &n);  
    int a[n];  
    printf("Enter %d elements: ", n);  
    for(i = 0; i < n; i++) scanf("%d", &a[i]);  
    printf("Enter position to delete: ");  
    scanf("%d", &pos);  
    for(i = pos; i < n-1; i++) a[i] = a[i+1];  
    printf("After deletion: ");  
    for(i = 0; i < n-1; i++) printf("%d ", a[i]);  
    printf("\n");  
    return 0;  
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog13  
Enter size: 4  
Enter 4 elements: 1 -4 -55 99  
Enter position to delete: 2  
After deletion: 1 -4 99  
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog13  
Enter size: 6  
Enter 6 elements: 1 2 3 60 67  
1  
Enter position to delete: 5  
After deletion: 1 2 3 60 67  
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog13  
Enter size: 10  
Enter 10 elements: 1 2 3 -99 -68 -699 -10 -20 -99 10393  
Enter position to delete: 10  
After deletion: 1 2 3 -99 -68 -699 -10 -20 -99  
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```

PROGRAM 14: Sort Array Elements in Ascending Order

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

#include <stdio.h>

```
int main() {
    int n, i, j, temp;
    printf("Enter size: ");
    scanf("%d", &n);
    int a[n];
    printf("Enter %d elements: ", n);
    for(i = 0; i < n; i++) scanf("%d", &a[i]);
    for(i = 0; i < n-1; i++) {
        for(j = i+1; j < n; j++) {
            if(a[i] > a[j]) {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
    }
    printf("Sorted: ");
    for(i = 0; i < n; i++) printf("%d ", a[i]);
    printf("\n");
    return 0;
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog14
Enter size: 4
Enter 4 elements: 12 10 -9 -99
Sorted: -99 -9 10 12
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog14
Enter size: 6
Enter 6 elements: 1 2 3 50 60 1
Sorted: 1 1 2 3 50 60
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog14
Enter size: 10
Enter 10 elements: -999 -998 997 996 1 2 3 4 67 69
Sorted: -999 -998 1 2 3 4 67 69 996 997
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```

PROGRAM 15: Merge Two Arrays

//Author: FRANCIS REUBEN R - RVUN25CSE408

// SECTION B - 51

```
#include <stdio.h>
```

```
int main() {
    int n1, n2, i;
    printf("Enter size of first array: ");
    scanf("%d", &n1);
    int a[n1];
    printf("Enter %d elements: ", n1);
    for(i = 0; i < n1; i++) scanf("%d", &a[i]);
    printf("Enter size of second array: ");
    scanf("%d", &n2);
    int b[n2], c[n1+n2];
    printf("Enter %d elements: ", n2);
    for(i = 0; i < n2; i++) scanf("%d", &b[i]);
    for(i = 0; i < n1; i++) c[i] = a[i];
    for(i = 0; i < n2; i++) c[n1+i] = b[i];
    printf("Merged: ");
    for(i = 0; i < n1+n2; i++) printf("%d ", c[i]);
    printf("\n");
    return 0;
}
```

OUTPUT:

```
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog15
Enter size of first array: 10
Enter 10 elements: 12 12 12 0 0 0 1 1 1 9
Enter size of second array: 10
Enter 10 elements: 24 24 24 1 1 1 0 0 0 18
Merged: 12 12 12 0 0 0 1 1 1 9 24 24 24 1 1 1 0 0 0 18
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog15
Enter size of first array: 5
Enter 5 elements: 1 2 3 4 5
Enter size of second array: 5
Enter 5 elements: 6 7 8 9 10
Merged: 1 2 3 4 5 6 7 8 9 10
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$ ./Prog15
Enter size of first array: 0
Enter 0 elements: Enter size of second array: 0
Enter 0 elements: Merged:
parallels@debian-gnu-linux-12:~/Documents/Frank/f_PC/SPLPROGS$
```